

SOKNEDAL, Norway

WMO: 012530

Lat: 62.953N

Lon: 10.179E

Elev: 299

StdP: 97.78

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-16.9	-21.5	0.6	-19.0	-19.1	0.7	-16.5	8.5	-1.9	7.3	-2.4	1.2	160	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	24.8	17.0	22.7	15.7	20.9	15.0	18.0	22.9	16.7	21.2	15.6	19.8	2.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.7	20.5	14.7	10.8	18.9	13.6	10.1	17.5	51.8	22.7	48.0	21.4	44.7	20.0	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.2	5.3	4.4	-20.9	28.5	3.7	2.5	-23.6	30.2	-25.8	31.7	-27.9	33.0	-30.6	34.8
			-21.2	19.9	3.7	1.5	-23.8	21.0	-26.0	21.8	-28.0	22.7	-30.7	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-4.9	-3.4	-0.6	3.1	8.0	11.0	14.2	13.1	9.8	3.9	-1.2	-3.7	
	DBStd	7.84	5.40	5.06	4.11	3.41	4.41	3.32	3.46	2.87	3.06	3.40	5.16	5.43	
	HDD10.0	2496	463	374	328	207	95	27	4	6	42	191	336	424	
	HDD18.3	5184	721	608	586	456	321	220	136	165	257	447	586	683	
	CDD10.0	363	0	0	0	2	32	57	134	101	35	3	0	0	
	CDD18.3	9	0	0	0	0	0	1	8	1	0	0	0	0	
	CDH23.3	147	0	0	0	0	11	14	108	14	0	0	0	0	
	CDH26.7	28	0	0	0	0	0	1	27	1	0	0	0	0	
Wind	WSAvg	1.8	2.1	1.9	2.0	1.8	2.0	1.8	1.6	1.5	1.6	1.7	1.7	1.8	
Precipitation	PrecAvg	821	65	62	59	43	51	76	86	86	80	70	68	72	
	PrecMax	1036	173	111	139	141	84	150	134	204	189	164	200	165	
	PrecMin	611	0	0	19	15	21	29	44	22	28	23	2	14	
	PrecStd	115	46	31	39	29	18	30	31	45	46	31	61	38	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	7.4	9.7	18.1	24.9	25.3	29.9	25.2	20.7	15.9	9.5	6.2	
		MCWB	3.8	4.2	5.3	10.5	15.2	16.2	19.1	17.4	14.4	12.6	7.3	3.9	
	2%	DB	4.2	5.6	7.6	14.5	22.2	22.0	26.5	22.8	18.7	12.7	7.9	5.1	
		MCWB	2.5	2.9	4.3	7.9	13.8	15.0	18.5	16.0	13.9	9.5	5.8	2.9	
	5%	DB	3.2	4.2	6.2	11.2	19.9	19.7	23.6	20.8	16.9	10.8	6.3	4.0	
		MCWB	1.7	2.0	3.4	5.9	12.8	13.9	16.8	15.4	13.1	8.3	4.5	2.1	
	10%	DB	2.0	3.0	4.9	8.8	16.7	17.6	21.3	18.9	15.1	8.9	5.0	2.8	
		MCWB	0.8	1.0	2.3	4.8	11.4	13.0	15.8	14.6	12.0	6.5	3.2	1.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.6	6.2	11.0	16.4	17.3	20.8	18.5	15.8	13.6	7.6	4.7
			MCDB	5.5	6.9	8.7	18.0	22.8	22.4	28.7	23.3	19.3	15.6	9.1	5.6
2%		WB	2.9	3.3	4.9	8.3	14.7	15.8	18.9	17.0	14.5	9.9	6.1	3.4	
		MCDB	3.8	5.1	7.1	13.8	20.8	20.8	25.3	20.9	17.7	12.3	7.6	4.5	
5%		WB	1.9	2.3	3.7	6.6	13.1	14.5	17.5	16.0	13.4	8.4	4.8	2.3	
		MCDB	3.0	3.9	5.9	10.5	19.1	19.0	22.5	19.8	16.5	10.6	6.2	3.5	
10%		WB	0.8	1.2	2.6	5.1	11.5	13.2	16.2	14.9	12.2	6.8	3.4	1.4	
		MCDB	1.8	2.8	4.5	8.4	16.6	17.0	20.6	18.4	14.8	8.5	4.7	2.5	
Mean Daily Temperature Range			MDBR	5.8	6.6	8.2	9.4	10.2	9.4	10.0	9.9	9.2	7.3	5.7	5.8
		5% DB	MCDBR	5.4	6.8	8.6	13.2	16.0	13.8	14.9	13.1	11.6	9.1	6.6	6.1
	MCWBR		4.3	4.9	5.6	7.6	8.6	7.6	7.8	7.3	7.3	6.4	5.1	5.0	
	5% WB	MCDBR	5.3	6.8	7.8	11.5	15.2	12.9	13.8	11.7	11.5	8.7	6.8	6.1	
		MCWBR	4.3	5.1	5.4	7.0	8.5	7.4	7.5	6.9	7.4	6.4	5.4	5.1	
Clear-Sky Solar Irradiance	taub	0.209	0.268	0.300	0.329	0.338	0.339	0.350	0.337	0.310	0.290	0.223	N/A		
	taud	2.041	2.219	2.229	2.288	2.407	2.463	2.457	2.486	2.546	2.455	2.118	N/A		
	Ebn at Noon	444	683	798	853	878	884	862	843	804	668	419	N/A		
	Edh at Noon	31	60	88	104	101	97	96	85	66	49	28	N/A		
All-Sky Solar Radiation	RadAvg	0.19	0.78	2.15	3.98	4.83	4.90	4.62	3.57	2.19	0.97	0.28	0.10		
	RadStd	0.03	0.11	0.24	0.28	0.29	0.43	0.46	0.36	0.22	0.12	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-143	-189	+67	N/A

Nomenclature: See separate page